National University of Computer and Emerging Sciences

FAST School of Computing | **Fall-2023** | **Islamabad Campus**

EL - 1005: Digital Logic Design Lab

BDS (A,B,C,D)

Thursday, 11th January, 2024

Lab Instructors

Fakhar Abbas, Sana Ehsan

Serial No:

Final Exam

Total Time: 3 Hours

Total Marks: 60

Signature of Invigilator

Student Name | **Roll No.** | **Course Section** | **Student Signature**

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work.
3. Verify that you have **eight (8)** different printed pages including this title page. There are **three (3)** questions.
4. Calculator sharing is strictly prohibited.
5. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.
6. Ensure that you do not have any electronic gadget (like mobile phone, smart watch, etc.) with you.

	Q-1	Q-2	Q-3	Total
Marks Obtained				
Total Marks	**20**	**25**	**15**	**60**

Vetter Signature & Date:

Question 1 [10+10 Marks]

Given the following function:

$$\boldsymbol{F}(\boldsymbol{A},\boldsymbol{B},\boldsymbol{C},\boldsymbol{D}) = \sum(\mathbf{1},\mathbf{2},\mathbf{5},\mathbf{7},\mathbf{8},\mathbf{10},\mathbf{11},\mathbf{13},\mathbf{15})$$

a) Implement it with a 4 x 1 MUX on Proteus. Don’t use 8 X1 MUX directly but you can build it using 4 x 1 multiplexers if required. You may use external gates if needed. Clearly show your working here. **[50% marks for Theoretical Work]**

b) Implement it with decoder. The choice of decoder is up to you. Clearly show your working here. **[50% marks for Theoretical Work]**

Question 2 [15 + 10 = 25 Marks]

a) Design a Combinational circuit that counts number of zeros in a 4-bit input number. Display the output on 7-Segment Digital display unit. Show your working below. **[50% marks for Theoretical Work]**

[Test Cases: If

- **i.** input: **0000, then 4** would be displayed on 7-Segment
- **ii.** input: **1100, then 2** would be displayed on 7-Segment
- **iii.** input: **1101, then 1** would be displayed on 7-Segment
- **iv.** input: **1111, then 0** would be displayed on 7-Segment
- **v.** input: **1000, then 3** would be displayed on 7-Segment

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b)

Given the following function:

$$F(A,B,C,D) = (A \oplus B)'(C \oplus D)$$

i. Implement it on Proteus using **NAND** Gates only **(No marks for Paper work)**

ii. Implement it on Proteus using **NOR** Gates only **(No marks for Paper work)**

Question 3 [20 Marks]

Design a Combinational circuit that perform the following functions on the 4-bit input number. Cleary show your working below **[50% marks for Theoretical Work]**

[Conditions: If

i. If **input** is even number like 0010 which is 2 in decimal**, then output** will be **1 more** than the input

ii. If **input** is odd number like 0011 which is 3 in decimal**, then output** will be **1 less** than the input

iii. If **input** is prime number like 0101 which is 5 in decimal, **then output** will be **2 more** than the input

Important Note: - If a number is both odd and prime like 3,5,7 etc. then perform the operation described in **iii-**
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Note: - The choice of implementation of this problem is your whichever method you adopt will be acceptable.

IC names and their numbers

1. Multiplexer (dual 74LS153 4-to 1)
2. Decoder: 74LS138 (3-to-8 Decoder)
3. Decoder (74154 4-to-16 Decoder)
4. 7SEG-BCD-GRN (Green, 7-Segment BCD Display)